Which note should ScentSelf use as its first test anchor?

Palaloy | MADT7202 | Code, notebook & results: Pornphrom/MADT7202_midterm

1 The decision

Palaloy, ScentSelf founder, will select a test anchor. A stable bridge that outranks prevalence can support differentiation; otherwise use only a provisional control, with no positioning claim.

2 Why a network

Frequency ignores paths; ordinary feature clustering groups similar notes but does not measure their role on paths between groups. Weighted betweenness measures that bridging role; prevalence ranks provide the comparison.

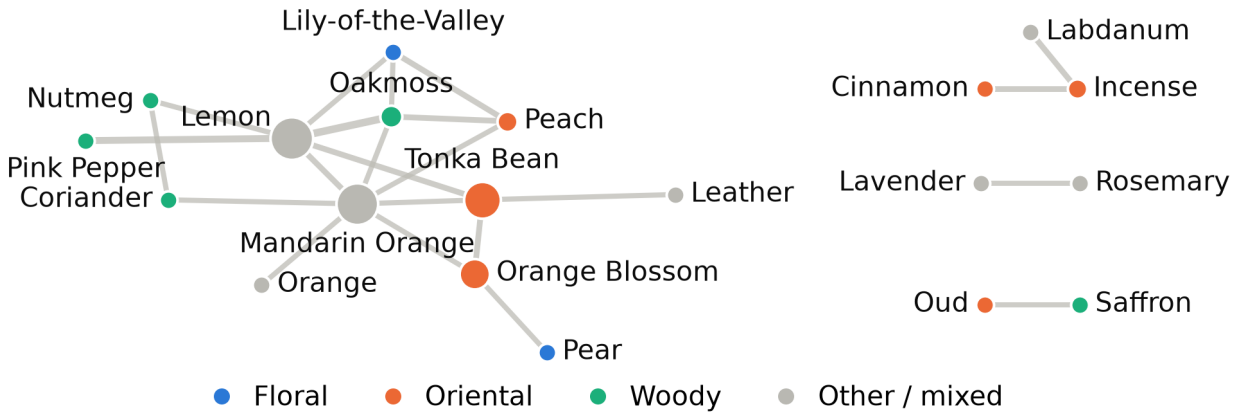
3 Data

Sources	Fragrantica (Selenium); Parfumo fills notes for **11 of 209 perfumes**. Parsed offline.
Period	30 Aug-12 Oct 2026.
Volume	**209** perfumes, 356 notes; wider set: **1,780**. URL manifest in repo.
Access	Fragrantica: no written permission; ToS 5.3 restricts automation. Parfumo: private/non-commercial copies allowed; project permission unconfirmed.
Sampling	Purposive list; Floral 61 vs Leather 4. Not a market sample; supplements are not triangulation.

4 Graph construction

Nodes / edges	Note / same-perfume co-occurrence. No seed notes: curated-list corpus.
Weight	Count favours repeated pairings; distance = 1/count. NPMI >0.1 checks prevalence.
Pruning	Before ranking: exclude generic/high-prevalence >=20% (10 notes); keep frequency >=5 (89), edges >=6; remove isolates.
Choice	k=6: nearest density 0.12 among k=3..10; **20 nodes / 23 edges**. k=8 leaves 3 nodes / 2 edges.
Rule	Post-result amendment (13 Oct; README): same-node average ranks; gap >=5, positive top-10 in >=80% of 300 brand resamples and at k=3,6,8.

5 The network



Node size = betweenness; colour = associated scent family; line width = pair count.

6 Findings

Note	Freq. rank	BC rank	Gap	Top-10 rate
Lemon	1.5	1	0.5	87.3%
Peach	12.5	6	6.5	25.3%

1. Peach has a 6.5-rank advantage but only 25.3% stability. Lemon is stable (87.3%) with little rank advantage. Neither qualifies.

Check	Pass	Control
Raw (23 edges)	0	Lemon
NPMI (22 edges)	0	Lemon
No supplements (198)	0	Lemon
Wider (1,780)	0	Orange Blossom

2. Zero anchors pass all criteria in each check; the wider catalogue changes the control. Keep Lemon specific to the curated list.

7 Recommended action (proposed)

13 Oct - Palaloy: select Lemon as provisional control; make no differentiation claim.

By 23 Oct: brief and cost matched-base Lemon vs Lemon + Peach prototypes; confirm tester availability.

By 30 Oct, if ready: blinded, order-balanced pilot with 20 testers; advance Peach only if at least 15 prefer it, otherwise retain control. Section 8 gates testing.

8 What would change my mind

Declared-note links do not prove liking. Permission is unresolved for both sites: obtain it or rerun permitted data before testing, with no assumed turnaround. A changed control in that rerun or a failed pilot reverses the recommendation.